WEEK 3 - INTRODUCTION TO MACHINE LEARNING

Student: Sahil Nabi

Course: Artificial Intelligence

Topic: Introduction to Machine Learning

Week: 3

Task 1: Machine Learning Fundamentals

1. What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every individual task. Instead of relying entirely on predefined rules, machine learning algorithms identify relationships within historical data and use those relationships to make predictions on new data. For example, a machine learning system can analyze information about students, such as study hours, attendance, and previous marks, and predict whether a student is likely to pass or fail.

2. AI vs Machine Learning vs Deep Learning

Artificial Intelligence (AI) is the broadest concept. It refers to systems that can perform tasks that normally require human intelligence, such as reasoning, decision-making, speech recognition, and problem-solving.

Machine Learning (ML) is a subset of AI. It allows systems to learn from data and improve their performance without requiring every rule to be manually programmed.

Deep Learning (DL) is a subset of Machine Learning that uses artificial neural networks with multiple layers. It is particularly useful for complex tasks involving images, speech, natural language, and large datasets.

Relationship: Artificial Intelligence → Machine Learning → Deep Learning

3. Dataset, Features and Labels

A dataset is a collection of information used for analysis or machine learning. Features are input variables used by a model to make predictions. In this project they are Study Hours, Attendance, Previous Marks, Assignment Score, and Project Score. A label or target is the output the model predicts; here it is Result, either PASS or FAIL.

4. Training Data and Testing Data

Training data is used to teach the machine learning model by identifying patterns and relationships between features and the target. Testing data is kept separate and is used to evaluate how well the trained model performs on previously unseen data. This project uses an 80:20 ratio.

5. Supervised Learning

Supervised Learning uses labelled data in which both input features and correct target values are provided during training. Examples include spam detection, student pass/fail prediction, house-price prediction, and fraud detection.

6. Unsupervised Learning

Unsupervised Learning works with data where the target or correct output is not provided. The algorithm attempts to discover hidden patterns or structures. Examples include customer segmentation, product grouping, anomaly identification, and market segmentation. K-Means is a common clustering algorithm.

7. Classification vs Regression

Classification predicts categories such as PASS/FAIL, spam/not spam, or fraud/not fraud. **Regression** predicts continuous numerical values such as house prices, temperature, sales revenue, or a student's numerical final score. This project is a classification problem.

8. Real-World Applications

• Healthcare: disease detection and medical image analysis.

• Banking: fraud detection, credit scoring, and risk analysis.

• E-commerce: recommendation systems.

• Transportation: route optimization and traffic prediction.

• Cybersecurity: suspicious-activity and malware detection.

• Education: student-performance analysis and early support.

Task 1 Conclusion: Machine Learning provides a way for computers to learn from data and make useful predictions. Understanding datasets, features, targets, training data, testing data, supervised learning, unsupervised learning, classification, and regression provides the foundation required to develop practical ML applications.

Task 2: Dataset Preparation

Dataset Used

A Student Performance dataset was created for this project containing 50 student records.

Feature	Description
Study_Hours	Number of hours studied
Attendance	Student attendance percentage
Previous_Marks	Previous academic marks
Assignment_Score	Assignment performance
Project_Score	Project performance
Result	PASS or FAIL

The dataset was created using Pandas and saved as `Dataset.csv`. It was loaded using `pd.read_csv("Dataset.csv")`. The first five rows were displayed using `head()`, the shape was checked using `shape`, and data types were examined using `dtypes`. Missing values were checked using `df.isnull().sum()`; none were found. The five numerical columns were selected as features and Result as the target.

Task 3: Train-Test Split and Data Preprocessing

The dataset was divided into training and testing sets using Scikit-learn's `train_test_split()`. An 80:20 split was used, with approximately 80% training data and 20% testing data. A random_state of 42 was used to make the split reproducible.

The target variable containing PASS and FAIL was converted into numerical values using LabelEncoder: FAIL = 0 and PASS = 1.

For preprocessing, StandardScaler was used to standardize the numerical features. The scaler was fitted only on the training data and then applied to both training and testing data. This prevents data leakage from the testing dataset.

Workflow: Dataset → Preprocessing → Train-Test Split → Model Training → Testing → Prediction → Evaluation

Task 4: First Machine Learning Model

A Logistic Regression model from Scikit-learn was selected for the classification problem. The model was trained using the preprocessed training data with `model.fit(X_train_scaled, y_train)`.

After training, predictions were generated using the testing data. The predicted results were compared with the actual results, and model accuracy was calculated using Scikit-learn's `accuracy_score()`.

A classification report was also generated to examine precision, recall, F1-score, and support for the PASS and FAIL classes. Logistic Regression was suitable because the target variable contains two classes: PASS and FAIL.

Task 4 Conclusion: The model successfully demonstrated the process of training a supervised classification model, making predictions on unseen data, and evaluating performance.